THE HOMOTOPY CATEGORY OF DIFFERENTIAL GRADED MODULES OVER A DIFFERENTIAL GRADED ALGEBRA IS PRE-TRIANGULATED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the TR1 verification from admissible sequences and cone triangles, the two-direction TR2 argument with the rotation isomorphisms and sign adjustments, and the TR3 reduction through lemma-functorial-cone are all retained verbatim.

Editorial note. The label lemma-homotopy-category-pre-triangulated names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Cone triangles as distinguished triangles. Same chapter, Lemma lemma-the-same-up-to-isomorphisms. Identifies distinguished triangles with cone triangles of homomorphisms up to isomorphism of triangles; every TR verification normalizes to this form.
- Rotated triangles and the rotated cone. Same chapter, Lemmas lemma-rotate-triangle and lemma-rotate-cone. Give the rotated triangle isomorphisms and the distinguishability of the rotated cone triangle, closing both directions of TR2.

Source reference. Original reference 014F: sources/stacks-dga/pinned-original-source.tex, source lines 644–644. Prerequisite pointer invoked at proof line(s) 1278; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 014O: sources/stacks-dga/pinned-original-source.tex, source lines 963–963. Prerequisite pointer invoked at proof line(s) 1268; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 014N: sources/stacks-dga/pinned-original-source.tex, source lines 979–979. Prerequisite pointer invoked at proof line(s) 1250; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 014L: sources/stacks-dga/pinned-original-source.tex, source lines 1070–1070. Prerequisite pointer invoked at proof line(s) 1239; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Lemma 0.1. *Let (A, d) be a differential graded algebra. The homotopy category $K(Mod_{(A,d)})$ with its natural translation functors and distinguished triangles is a pre-triangulated category.*

Proof. Proof of TR1. By definition every triangle isomorphic to a distinguished one is distinguished. Also, any triangle $(K, K, 0, 1, 0, 0)$ is distinguished since $0 \to K \to K \to 0 \to 0$ is an admissible short exact sequence. Finally, given any homomorphism $f : K \to L$ of differential graded A-modules the triangle $(K, L, C(f), f, i, -p)$ is distinguished by Lemma 014L.

Proof of TR2. Let (X, Y, Z, f, g, h) be a triangle. Assume $(Y, Z, X[1], g, h, -f[1])$ is distinguished. Then there exists an admissible short exact sequence $0 \to K \to L \to M \to 0$ such that the associated triangle $(K, L, M, \alpha, \beta, \delta)$ is isomorphic to $(Y, Z, X[1], g, h, -f[1])$. Rotating back we see that (X, Y, Z, f, g, h) is isomorphic to $(M[-1], K, L, -\delta[-1], \alpha, \beta)$. It follows from Lemma 014N that the triangle $(M[-1], K, L, \delta[-1], \alpha, \beta)$ is isomorphic to $(M[-1], K, C(\delta[-1]), \delta[-1], i, p)$. Precomposing the previous isomorphism of triangles with -1 on Y it follows that (X, Y, Z, f, g, h) is isomorphic to $(M[-1], K, C(\delta[-1]), \delta[-1], i, -p)$. Hence it is distinguished by Lemma 014L. On the other hand, suppose that (X, Y, Z, f, g, h) is distinguished. By Lemma 014L this means that it is isomorphic to a triangle of the form $(K, L, C(f), f, i, -p)$ for some morphism f of $\mathrm{Mod}_{(A,\mathrm{d})}$. Then the rotated triangle $(Y, Z, X[1], g, h, -f[1])$ is isomorphic to $(L, C(f), K[1], i, -p, -f[1])$ which is isomorphic to the triangle $(L, C(f), K[1], i, p, f[1])$. By Lemma 014O this triangle is distinguished. Hence $(Y, Z, X[1], g, h, -f[1])$ is distinguished as desired.

Proof of TR3. Let (X, Y, Z, f, g, h) and (X', Y', Z', f', g', h') be distinguished triangles of $K(\mathcal{A})$ and let $a : X \to X'$ and $b : Y \to Y'$ be morphisms such that $f' \circ a = b \circ f$. By Lemma 014L we may assume that $(X, Y, Z, f, g, h) = (X, Y, C(f), f, i, -p)$ and $(X', Y', Z', f', g', h') = (X', Y', C(f'), f', i', -p')$. At this point we simply apply Lemma 014F to the commutative diagram given by f, f', a, b. □